

Week 6 (CS 418 @ UIC)

Week 6 Slide Deck

Visualizations

Jack Bandy 2026



Demo Content Slide

Content for Week 6.

- Visualization refinement
- Interaction
- Time series
- Communicating uncertainty
- Sneak peek at modeling



Bubbles

FACTS

TEACH

ABOUT



HOW TO USE

Share

English

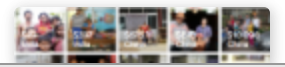
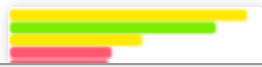


Life expectancy, at birth

GDP per capita, PPP (constant 2021 international \$)



FIND



embedded web page: [https://www.gapminder.org/tools/#\\$chart-type=bubbles&url=v2](https://www.gapminder.org/tools/#$chart-type=bubbles&url=v2)

Time Series Visualization

Time Series Basics

A **univariate time series** is simply a sequence of observations of the **same variable** collected over time - not just a “bag of values”

Two goals of time series analysis:

- **Understand** or explain processes that produced the data
- **Forecast** future values of the series (i.e. prediction)

Time Series Examples

- Airline passengers
- Public transit ridership
- Hourly/Daily/Monthly temperatures
- Annual spending
- Others you can think of?

Why I love time series

Time series can contextualize (almost) any arbitrary datum

- e.g. “UIC had XXXX graduates last year”
- e.g. “Chicago’s budget grew by \$XXmillion last year”
- e.g. “the mortgage interest rate is X.X%”
- e.g. “XX people have died from the flu this year”

Airline Passengers: Setup

```
1 import polars as pl
2 import seaborn as sns
3 import matplotlib.pyplot as plt
4
5 airline = pl.read_csv("../datasets/airline-passengers/airline-passengers.csv")
6 airline.sample(5, seed=42).sort("Month")
```

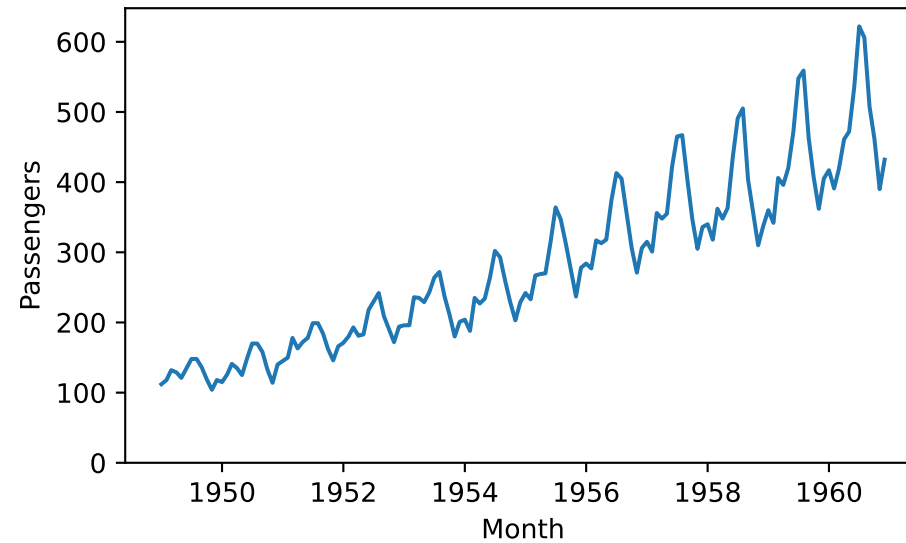
shape: (5, 2)

Month	Passengers
str	i64
"1952-09-01"	209
"1957-05-01"	355
"1958-07-01"	491
"1960-08-01"	606

"1960-12-01" 432

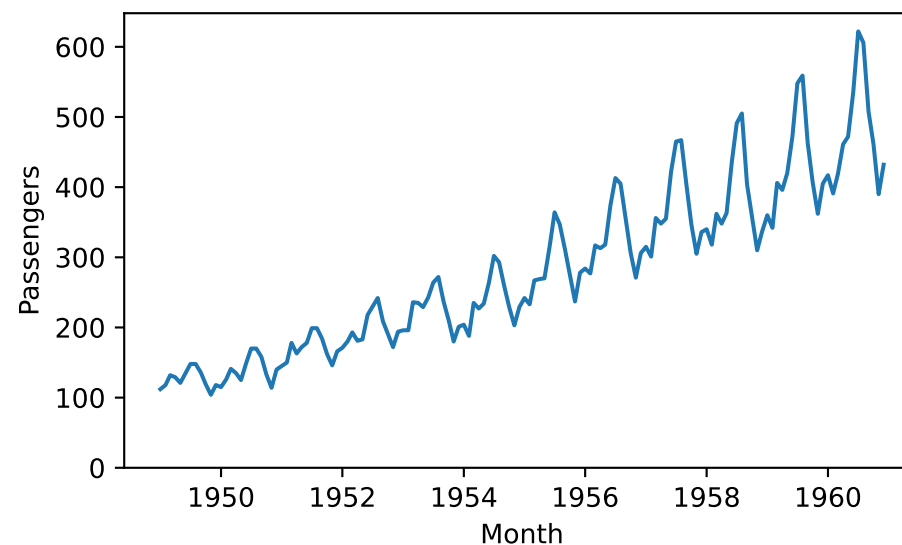
Airline Passengers Example

```
1 airline = airline.with_columns(pl.col("Month").str.to_date("%Y-%m-%d"))
2 ax = sns.lineplot(x=airline["Month"], y=airline["Passengers"])
3 ax.set_ylim(bottom=0)
4 plt.tight_layout()
```



Airline Passengers: What Do You Notice?

- How to describe this series?
- Overall trend(s)?
- Seasonal changes?
- Potential explanations?



Rolling Average

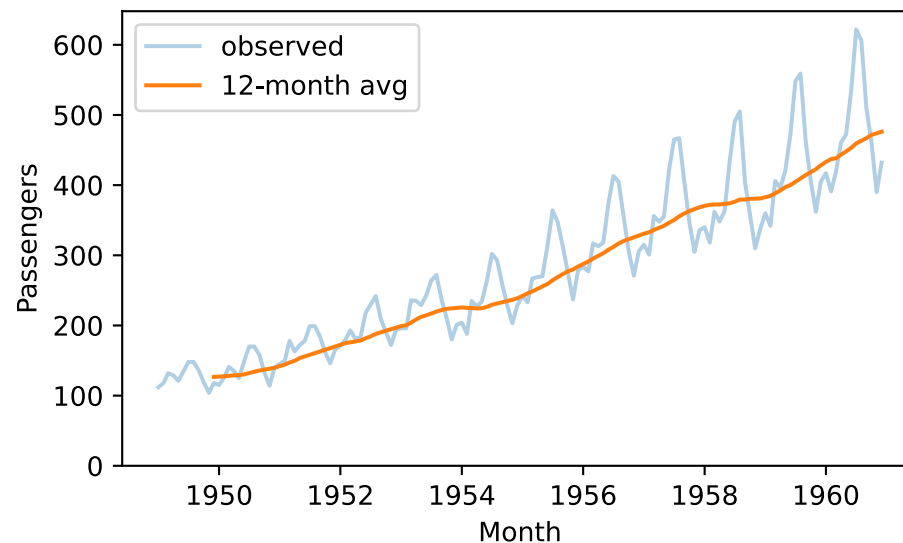
```
1 airline = airline.with_columns(  
2     pl.col("Passengers").rolling_mean(window_size=12).alias("trend")  
3 )  
4 airline.sample(5, seed=42).sort("Month")
```

shape: (5, 3)

Month	Passengers	trend
date	i64	f64
1952-09-01	209	190.0833333
1957-05-01	355	342.0833333
1958-07-01	491	376.3333333
1960-08-01	606	463.3333333
1960-12-01	432	476.1666667

Rolling Average

```
1 ax = sns.lineplot(x=airline["Month"], y=airline["Passengers"], alpha=0.35, label="observed")
2 trend_not_null = airline.filter(pl.col("trend").is_not_null())
3 sns.lineplot(x=trend_not_null["Month"], y=trend_not_null["trend"], label="12-month avg", ax=ax)
4 ax.legend()
5 ax.set_ylim(bottom=0)
6 plt.tight_layout()
```



Four-part Decomposition

Time series can be decomposed into four parts:

- **Level** — Baseline average value
- **Trend** — Long-term increase or decrease
- **Seasonality** — Repeating periodic pattern(s)
- **Noise** — Random/unexplained variation

Additive vs. Multiplicative Seasonality

Additive

$$y(t) = T + S + N$$

Seasonal swings are **constant**: regardless of trend level, variation is a fixed amount.

→ *Use when amplitudes stay flat.*

(Basically, if peaks get taller as the series continues, use multiplicative.)

Multiplicative

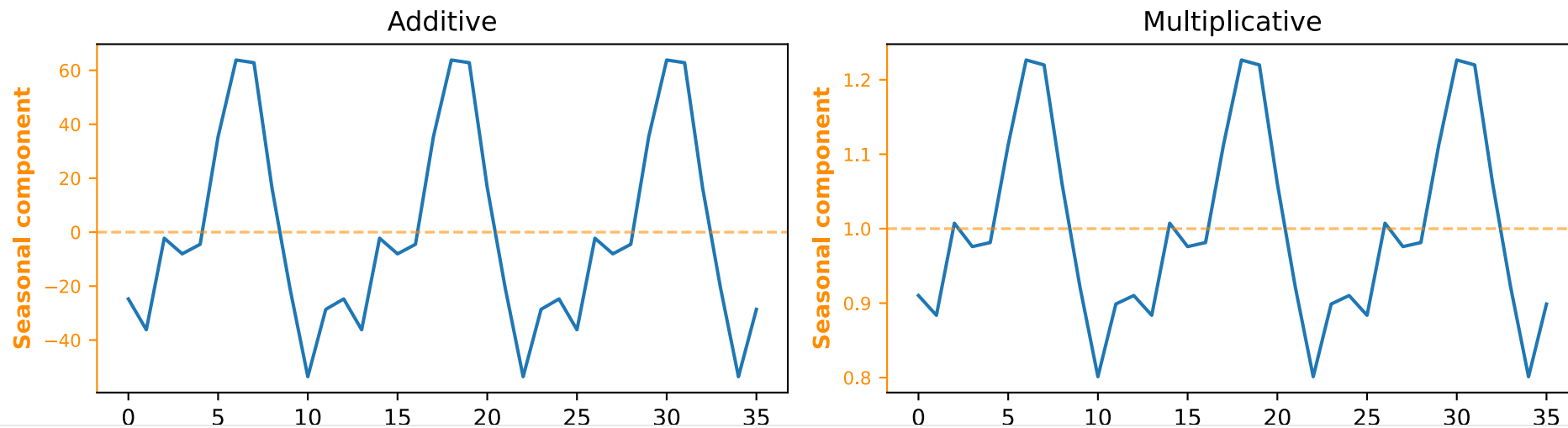
$$y(t) = T \times S \times N$$

Seasonality **scales as the trendline continues** (bigger baseline means bigger swings).

→ *Use when amplitudes grow.*

Additive vs. Multiplicative: Side by Side

```
1 from statsmodels.tsa.seasonal import seasonal_decompose
2 fig, axes = plt.subplots(1, 2, figsize=(10, 3.09))
3 for model, ax in zip(["additive", "multiplicative"], axes):
4     r = seasonal_decompose(airline["Passengers"].to_numpy().astype(float), model=model, period=12)
5     ax.plot(r.seasonal[:36])
6     ax.set_title(f"{model.capitalize()}")
7     ax.set_xlabel("Month (first 3 years)")
8     ax.set_ylabel("Seasonal component", color="darkorange", fontweight="bold")
9     ax.axhline(0 if model == "additive" else 1, color="darkorange", linewidth=1.2, linestyle="--", alpha=0.6)
10    ax.tick_params(axis="y", colors="darkorange", labels=8)
11    ax.spines["left"].set_color("darkorange")
12 plt.tight_layout()
```



Choosing the Right Seasonality Model

Use additive when seasonal swings are consistently the same size.

- *Example: a stadium closes outdoor seating during the winter, which reduces capacity by a fixed amount (e.g. 2,000 seats).*

Use multiplicative when seasonal swings scale with the trend.

- *Example: airline passengers — summer peaks are larger as the overall baseline gets higher (more people are flying).*

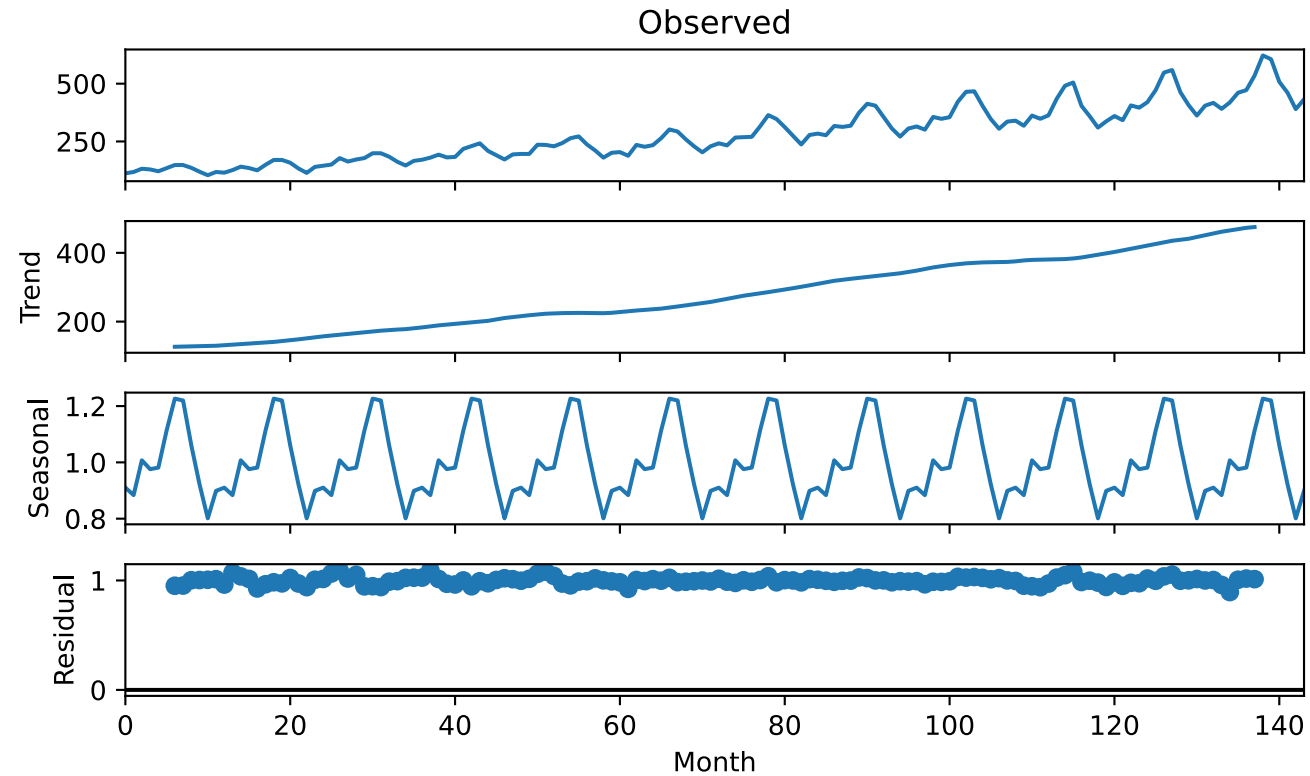
Remember to plot the raw data, check if the peaks get taller.

seasonal_decompose: Four Components

```
1 from statsmodels.tsa.seasonal import seasonal_decompose
2
3 values = airline["Passengers"].to_numpy().astype(float)
4 result = seasonal_decompose(values, model="multiplicative", period=12)
```

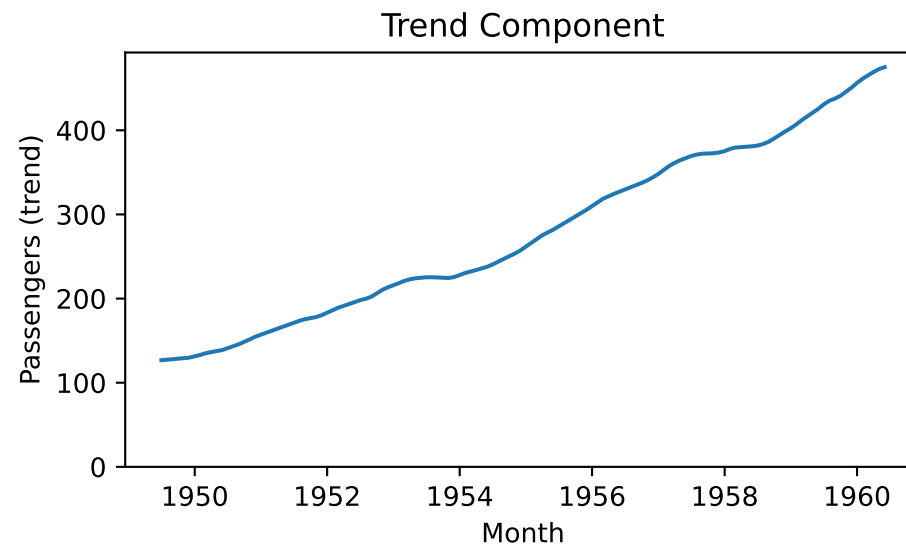
seasonal_decompose: Four Components

```
1 fig = result.plot()  
2 fig.set_size_inches(7, 4.3)  
3 fig.get_axes()[-1].set_xlabel("Month")  
4 plt.tight_layout()
```



The Trend Component

```
1 trend_df = pl.DataFrame({
2     "Month": airline["Month"],
3     "trend": pl.Series("trend", result.trend).fill_nan(None),
4 }).drop_nulls("trend")
5
6 ax = sns.lineplot(x=trend_df["Month"], y=trend_df["trend"])
7 ax.set_ylim(bottom=0)
8 ax.set_ylabel("Passengers (trend)")
9 ax.set_title("Trend Component")
10 plt.tight_layout()
```



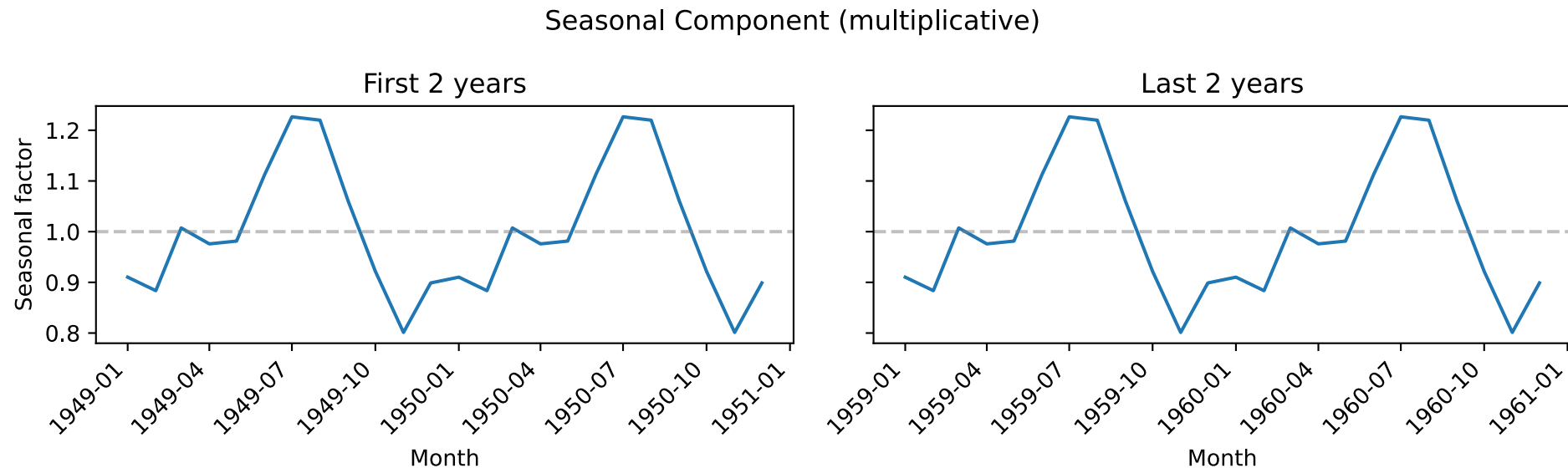
The Seasonal Component

```
1 seasonal_df = pl.DataFrame({
2     "Month": airline["Month"],
3     "seasonal": pl.Series("seasonal", result.seasonal),
4 })
5 first_two = seasonal_df.head(24)
6 last_two = seasonal_df.tail(24)
```

We'll check to see if the seasonal factor stays **constant** over time (with a multiplicative model)

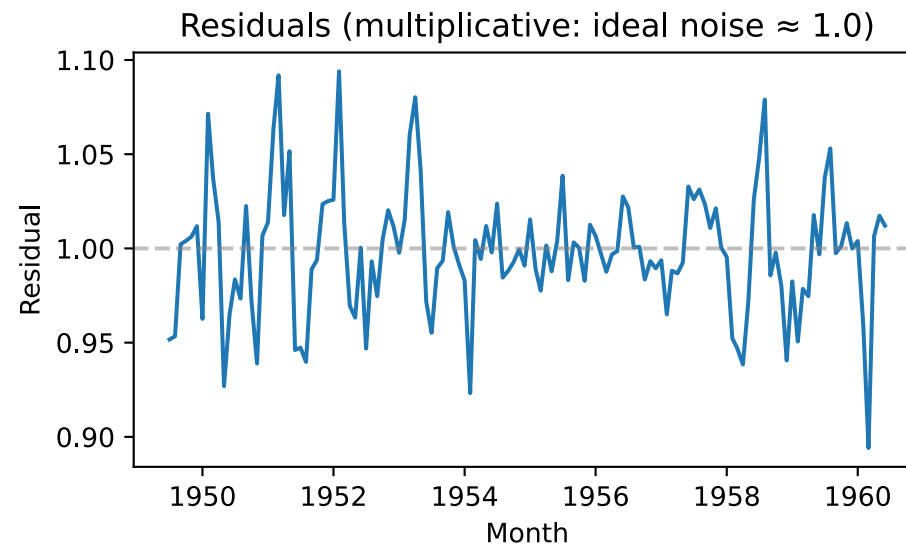
The Seasonal Component: Stable Over Time?

```
1 fig, axes = plt.subplots(1, 2, figsize=(10, 3.09), sharey=True)
2 for df, ax, label in zip([first_two, last_two], axes, ["First 2 years", "Last 2 years"]):
3     sns.lineplot(x=df["Month"], y=df["seasonal"], ax=ax)
4     ax.axhline(1.0, color="gray", linestyle="--", alpha=0.5)
5     ax.set(xlabel="Month", title=label)
6     plt.setp(ax.xaxis.get_majorticklabels(), rotation=45, ha="right")
7 axes[0].set_ylabel("Seasonal factor")
8 fig.suptitle("Seasonal Component (multiplicative)")
9 plt.tight_layout()
```



The Residuals

```
1 resid_df = pl.DataFrame({
2     "Month": airline["Month"],
3     "resid": pl.Series("resid", result.resid).fill_nan(None),
4 }).drop_nulls("resid")
5
6 ax = sns.lineplot(x=resid_df["Month"], y=resid_df["resid"])
7 ax.axhline(1.0, color="gray", linestyle="--", alpha=0.5)
8 ax.set_ylabel("Residual")
9 ax.set_title("Residuals (multiplicative: ideal noise  $\approx 1.0$ )")
10 plt.tight_layout()
```



Stationarity

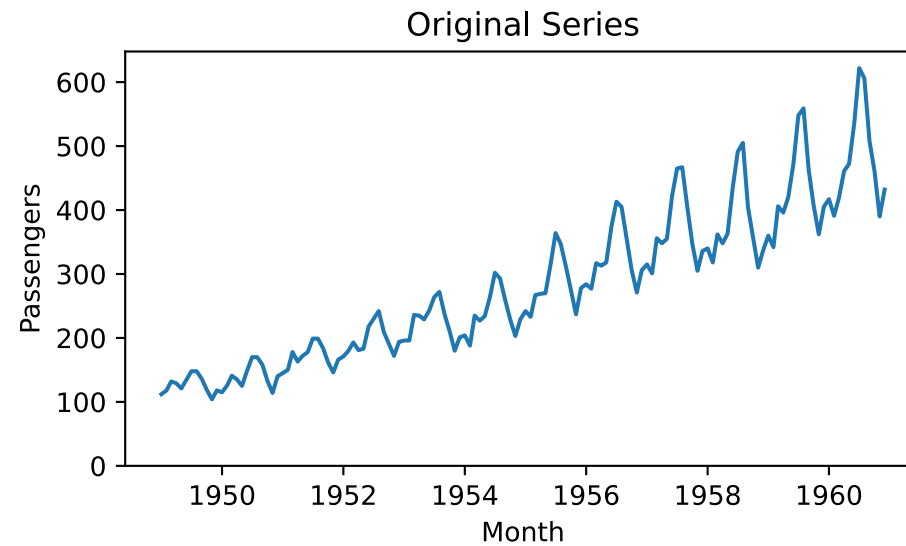
- Some forecasting models assume stationarity
- So you must know whether a series is stationary before modeling

A stationary series is defined by:

- **Constant mean** — no long-term drift up or down
- **Constant variance** — spread stays the same over time
- **No trend or seasonality**

Stationary?

```
1 airline_diff = airline.with_columns(  
2     pl.col("Passengers").diff().alias("first_diff")  
3 ).drop_nulls()  
4  
5 ax = sns.lineplot(x=airline["Month"], y=airline["Passengers"])  
6 ax.set_ylim(bottom=0)  
7 ax.set_ylabel("Passengers")  
8 ax.set_title("Original Series")  
9 plt.tight_layout()
```



Making a Series Stationary: Differencing

Differencing removes trend by computing the difference between consecutive observations:

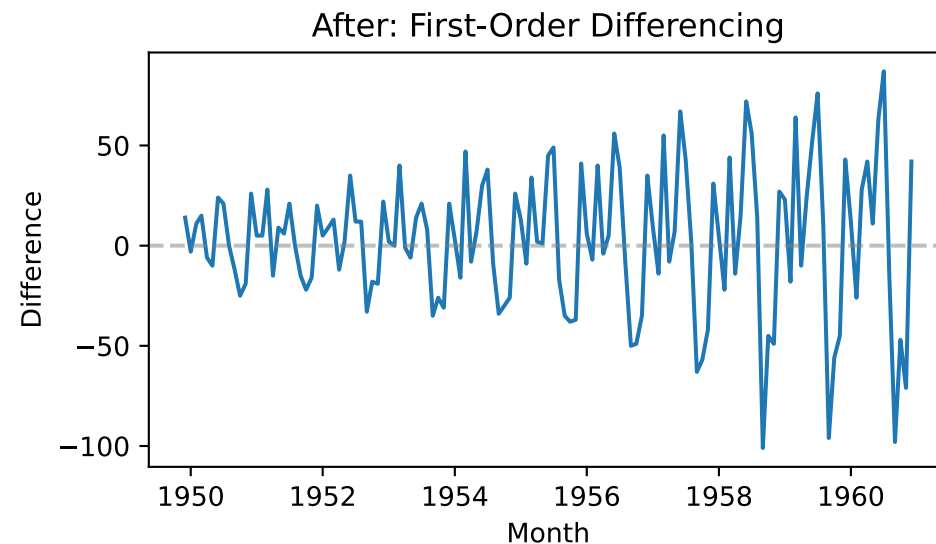
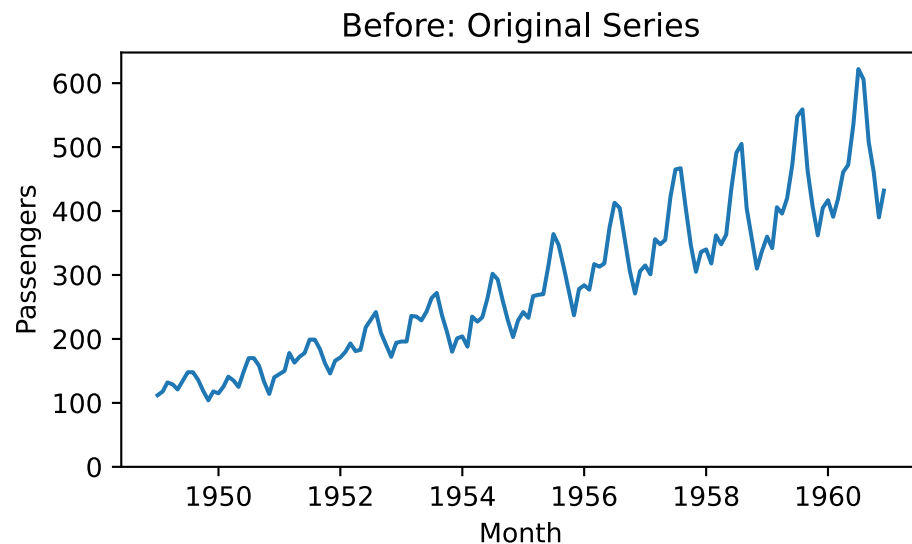
$$Y'_t = Y_t - Y_{t-1}$$

This is called **first-order differencing** (difference order = 1).

- (Recognize this concept?)
- Repeat the process for stronger trends
 - second-order differencing removes quadratic trends
- *Example: if passengers are $100 \rightarrow 120 \rightarrow 139$, the first differences are 20, 19 (much flatter)*

Differencing: Before and After

```
1 fig, axes = plt.subplots(1, 2, figsize=(10, 3.09))
2 sns.lineplot(x=airline["Month"], y=airline["Passengers"], ax=axes[0])
3 axes[0].set_ylim(bottom=0); axes[0].set_title("Before: Original Series")
4 sns.lineplot(x=airline_diff["Month"], y=airline_diff["first_diff"], ax=axes[1])
5 axes[1].axhline(0, color="gray", linestyle="--", alpha=0.5)
6 axes[1].set_ylabel("Difference"); axes[1].set_title("After: First-Order Differencing")
7 plt.tight_layout()
```



Time Series Example: CTA Rides

Load UIC-Halsted Daily Ridership

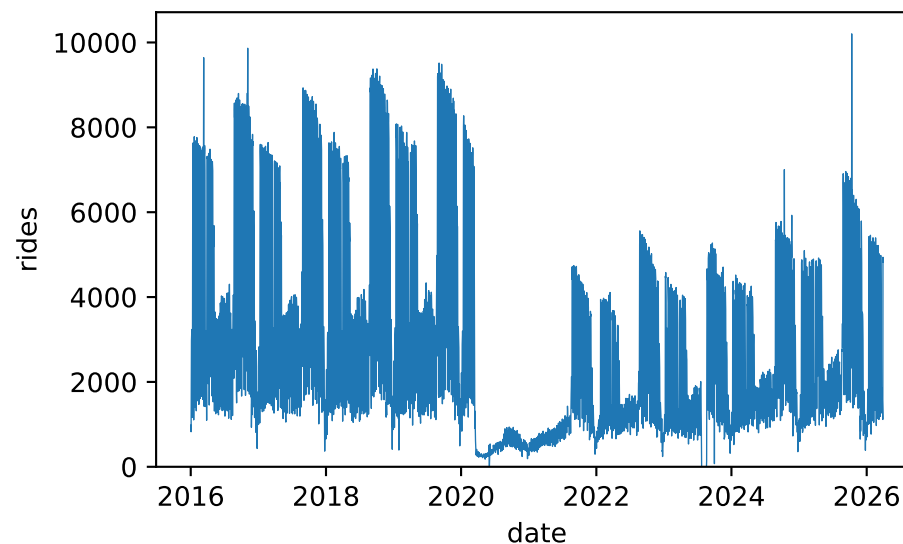
```
1 cta = pl.read_csv("../..../datasets/cta-ridership/Station_Entries_-_Daily_Totals_20260527.csv")
2 uic = cta.filter(pl.col("stationname") == "UIC-Halsted")
3 uic.sample(5, seed=42)
```

shape: (5, 5)

station_id	stationname	date	daytype	rides
i64	str	str	str	str
40350	"UIC-Halsted"	"07/19/2021"	"W"	"1,068"
40350	"UIC-Halsted"	"01/19/2009"	"W"	"1,824"
40350	"UIC-Halsted"	"11/01/2025"	"A"	"1,811"
40350	"UIC-Halsted"	"09/12/2018"	"W"	"9,375"
40350	"UIC-Halsted"	"01/12/2021"	"W"	"566"

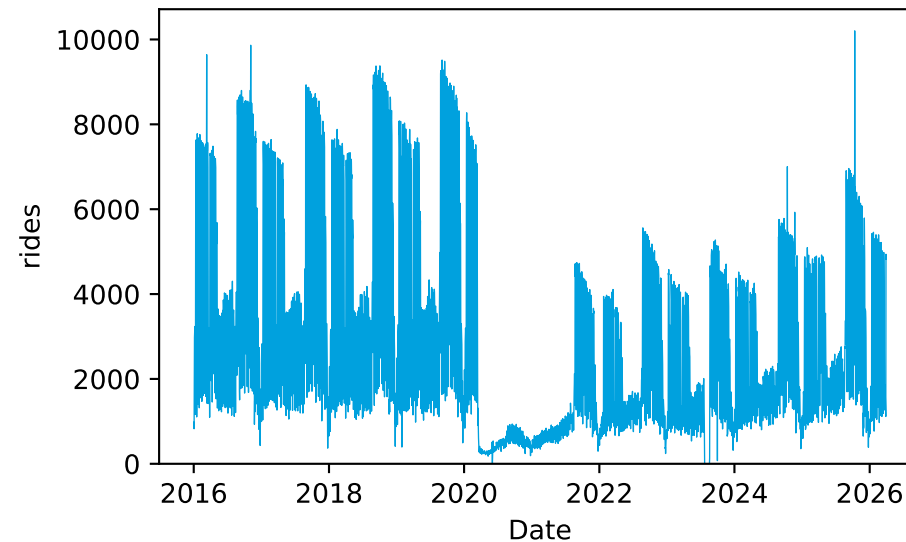
Load UIC-Halsted Daily Ridership

```
1 uic = (  
2     cta.filter(pl.col("stationname") == "UIC-Halsted")  
3     .with_columns(  
4         pl.col("date").str.to_date("%m/%d/%Y"),  
5         pl.col("rides").str.replace_all(",", "").cast(pl.Int64),  
6     ).sort("date").filter(pl.col("date") >= pl.date(2016, 1, 1))  
7 )  
8 ax = sns.lineplot(x=uic["date"], y=uic["rides"], linewidth=0.5)  
9 ax.set_ylim(bottom=0)  
10 plt.tight_layout()
```



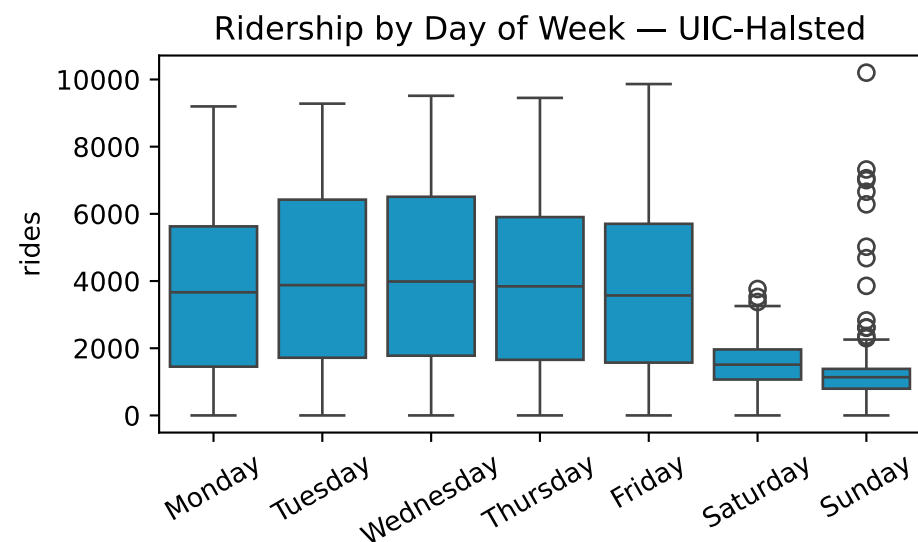
Just for Fun

```
1 CTA_BLUE = "#00A1DE"
2 ax = sns.lineplot(x=uic["date"], y=uic["rides"], linewidth=0.5, color=CTA_BLUE)
3 ax.set_xlabel("Date")
4 ax.set_ylim(bottom=0)
5 plt.tight_layout()
```



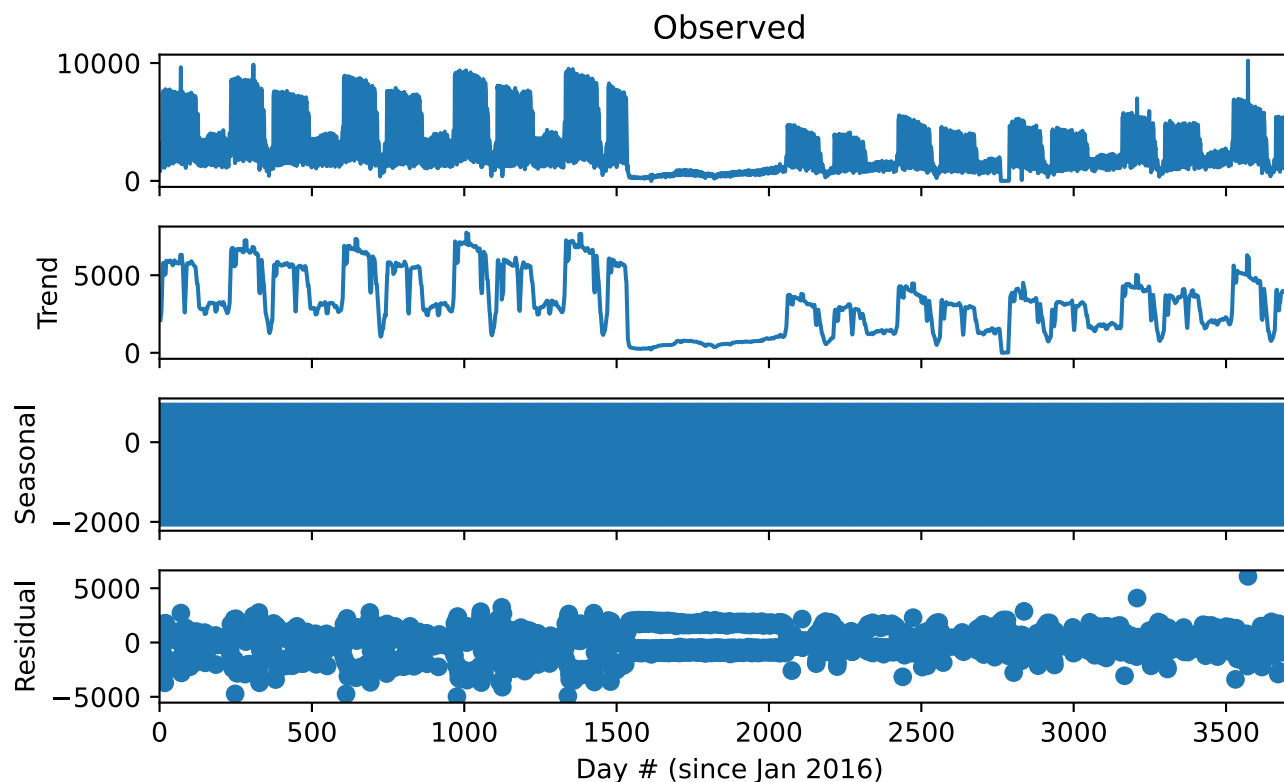
Weekday variation

```
1 day_order = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]
2 uic_dow = uic.with_columns(pl.col("date").dt.strftime("%A").alias("day_name"))
3 ax = sns.boxplot(x=uic_dow["day_name"], y=uic_dow["rides"], order=day_order, color=CTA_BLUE)
4 ax.set_xlabel("")
5 ax.tick_params(axis="x", rotation=30)
6 ax.set_title("Ridership by Day of Week — UIC-Halsted")
7 plt.tight_layout()
```



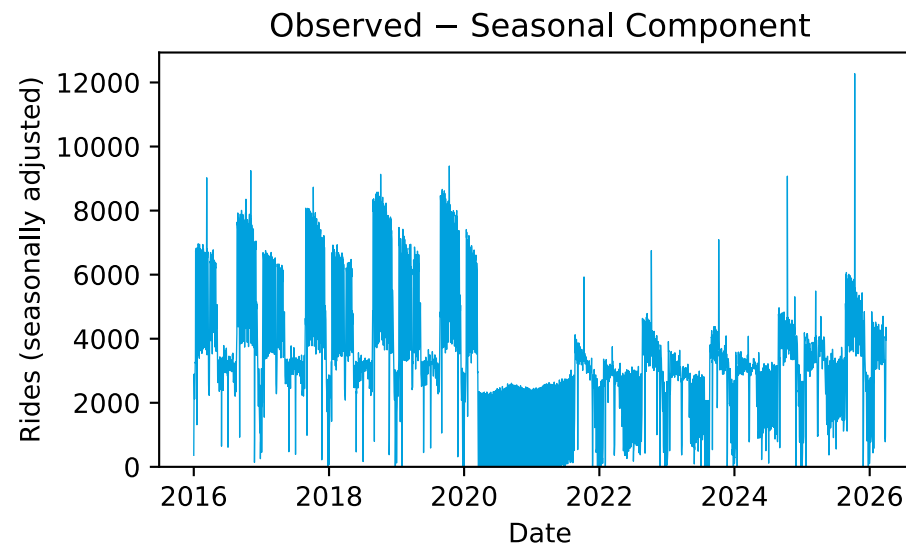
Decomposing with Period = 7

```
1 rides = uic["rides"].to_numpy().astype(float)
2 cta_result = seasonal_decompose(rides, model="additive", period=7)
3 fig = cta_result.plot()
4 fig.set_size_inches(7, 4.3)
5 fig.get_axes()[-1].set_xlabel("Day # (since Jan 2016)")
6 plt.tight_layout()
```



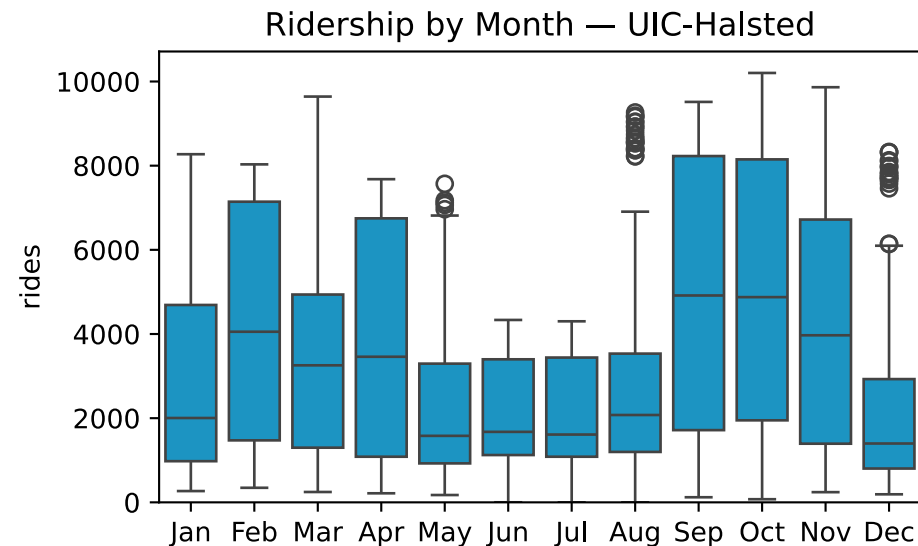
Period=7 Decomposition: What's Still There?

```
1 deseasonalized = pl.DataFrame({
2     "date": uic["date"],
3     "rides": pl.Series("rides", cta_result.observed - cta_result.seasonal),
4 })
5 ax = sns.lineplot(x=deseasonalized["date"], y=deseasonalized["rides"], color=CTA_BLUE, linewidth=0.5)
6 ax.set_xlabel("Date")
7 ax.set_ylabel("Rides (seasonally adjusted)")
8 ax.set_title("Observed - Seasonal Component")
9 ax.set_ylim(bottom=0)
10 plt.tight_layout()
```



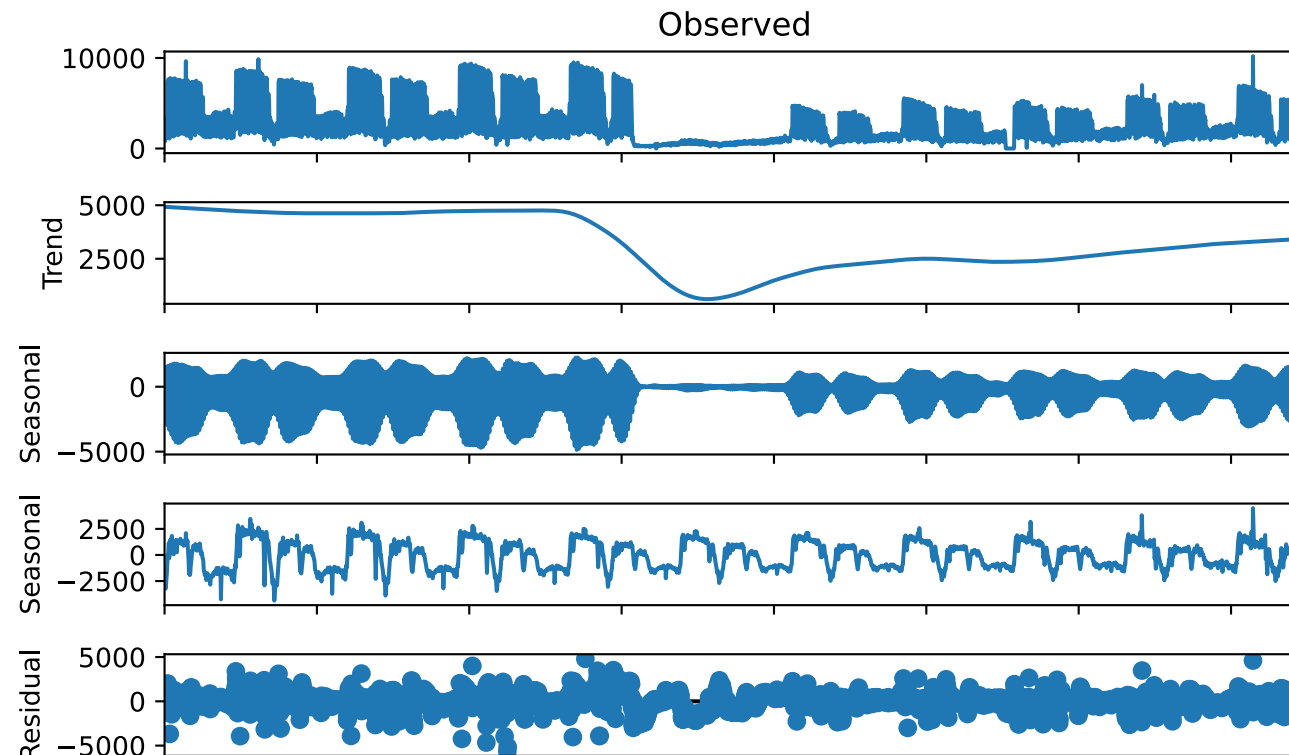
Monthly Variation(s) for UIC-Halsted Stop

```
1 month_order = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",  
2               "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]  
3 uic_month = uic.with_columns(pl.col("date").dt.strftime("%b").alias("month"))  
4 ax = sns.boxplot(x=uic_month["month"], y=uic_month["rides"], order=month_order, color=CTA_BLUE)  
5 ax.set_xlabel("")  
6 ax.set_ylim(bottom=0)  
7 ax.set_title("Ridership by Month — UIC-Halsted")  
8 plt.tight_layout()
```



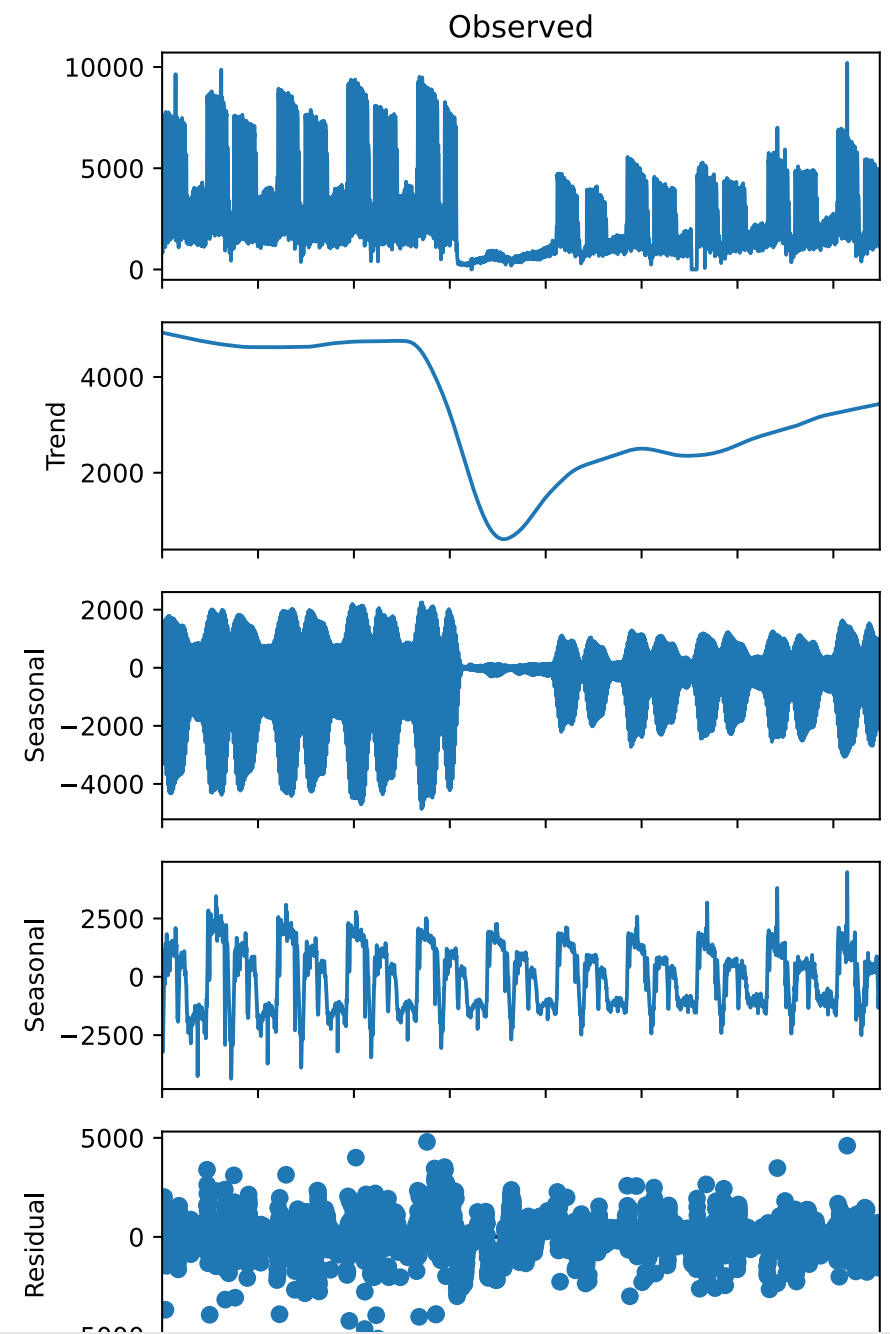
MSTL: Multiple Seasonal Decomposition

```
1 from statsmodels.tsa.seasonal import MSTL
2
3 mstl = MSTL(rides, periods=(7, 365)).fit()
4 fig = mstl.plot()
5 fig.set_size_inches(7, 4.6)
6 fig.get_axes()[-1].set_xlabel("Day # (since Jan 2016)")
7 plt.tight_layout()
```



Understanding the Decomposition

- **Trend:** steady through ~2019, collapse in spring 2020 (COVID), gradual recovery
- **Weekly seasonality:** weekday vs weekend ridership
- **Annual seasonality:** dips in January, summer, and December (academic calendar)
- **Residuals:** large spikes or dips mark unusual days — holidays,



A Different Station: Washington/Wabash

Load Washington/Wabash

```
1 CTA_ORANGE = "#F9461C"
2 wabash = (
3     cta.filter(pl.col("stationname") == "Washington/Wabash")
4     .with_columns(
5         pl.col("date").str.to_date("%m/%d/%Y"),
6         pl.col("rides").str.replace_all(",", "").cast(pl.Int64),
7     ).sort("date").filter(pl.col("date") >= pl.date(2016, 1, 1))
8 )
9 wabash.sample(5, seed=42).sort("date")
```

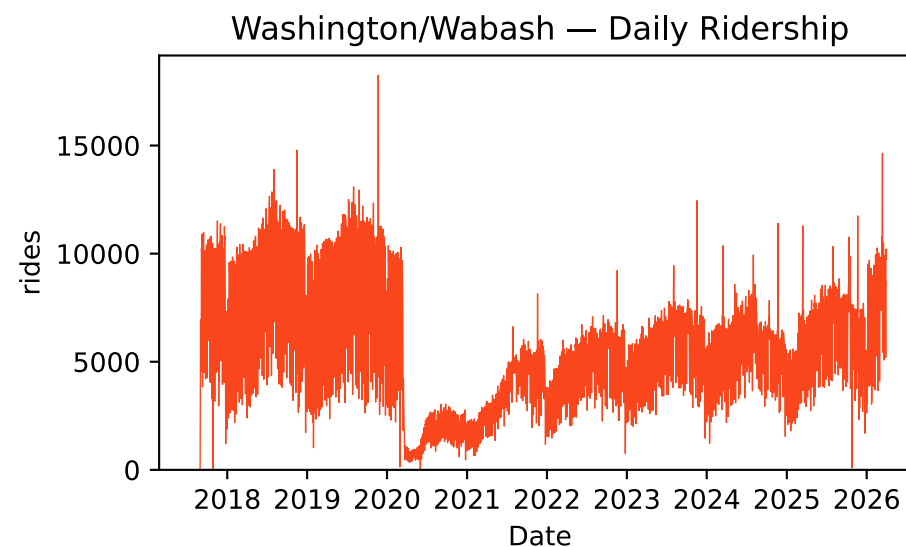
shape: (5, 5)

station_id	stationname	date	daytype	rides
i64	str	date	str	i64
41700	"Washington/Wabash"	2020-05-05	"W"	849
41700	"Washington/Wabash"	2023-08-28	"W"	6363
41700	"Washington/Wabash"	2024-06-16	"U"	4376

41700	"Washington/Wabash"	2024-08-18	"U"	4216
41700	"Washington/Wabash"	2026-02-07	"A"	5544

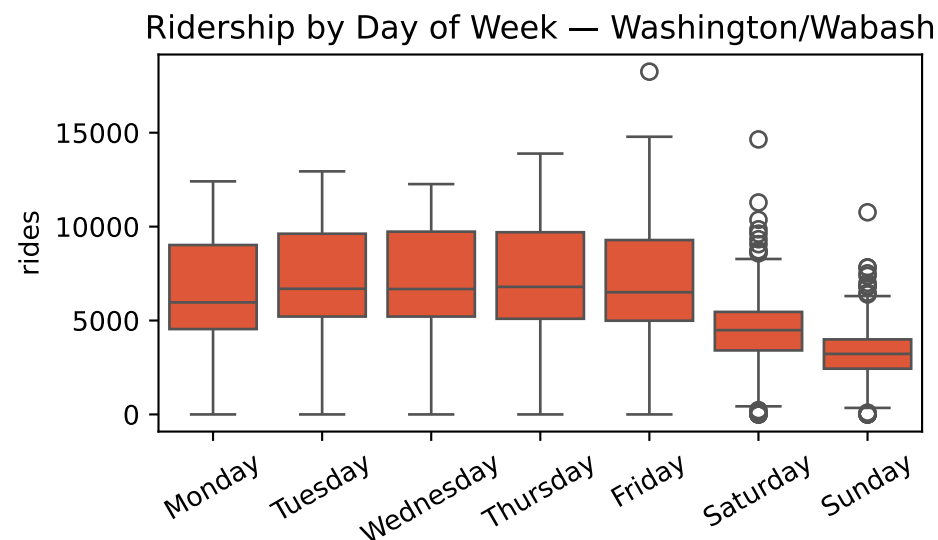
Washington/Wabash: Ridership Over Time

```
1 ax = sns.lineplot(x=wabash["date"], y=wabash["rides"], linewidth=0.5, color=CTA_ORANGE)
2 ax.set_xlabel("Date")
3 ax.set_ylim(bottom=0)
4 ax.set_title("Washington/Wabash — Daily Ridership")
5 plt.tight_layout()
```



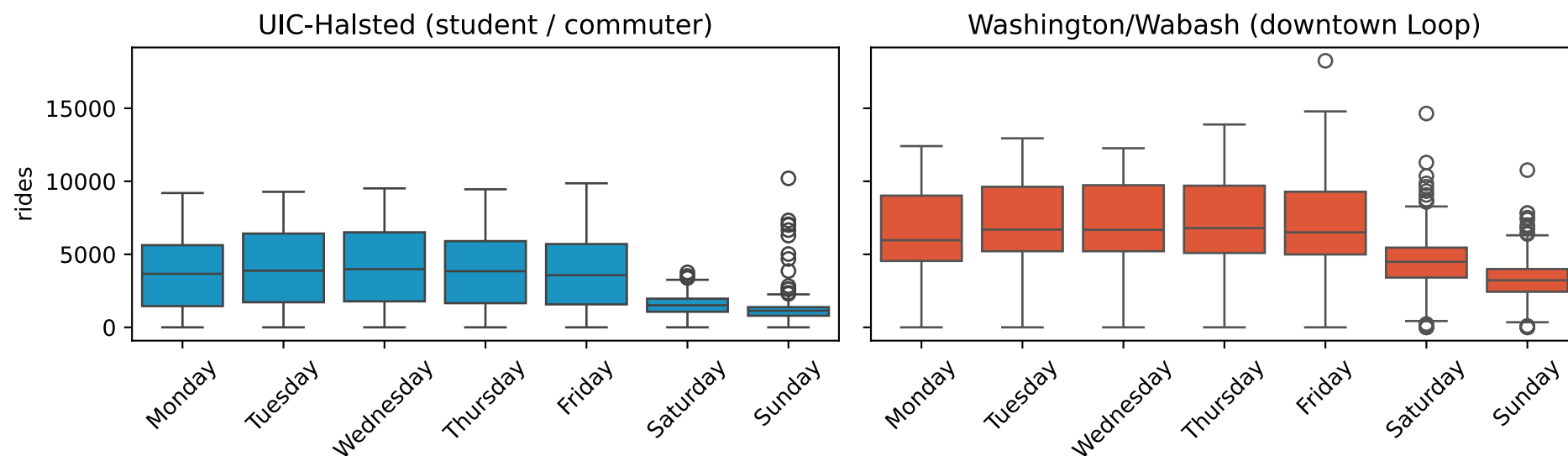
Washington/Wabash: Weekday Patterns

```
1 wabash_dow = wabash.with_columns(pl.col("date").dt.strftime("%A").alias("day_name"))
2 ax = sns.boxplot(x=wabash_dow["day_name"], y=wabash_dow["rides"], order=day_order, color=CTA_ORANGE)
3 ax.set_xlabel("")
4 ax.tick_params(axis="x", rotation=30)
5 ax.set_title("Ridership by Day of Week — Washington/Wabash")
6 plt.tight_layout()
```



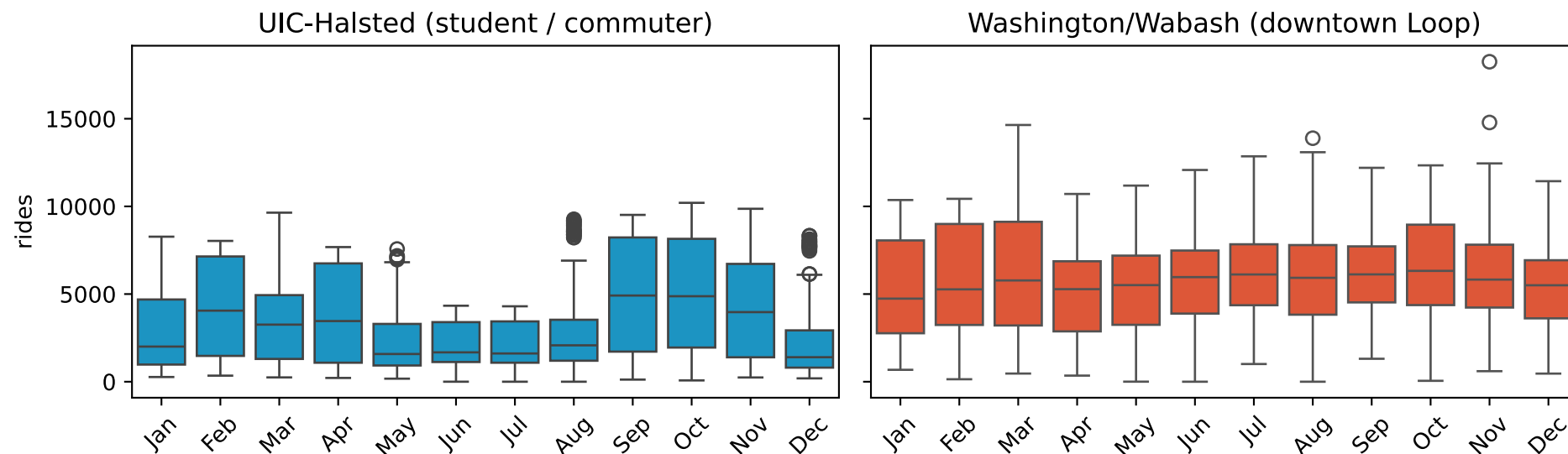
Two Stations, Two Rhythms? (Weekday)

```
1 fig, axes = plt.subplots(1, 2, figsize=(10, 3.09), sharey=True)
2 sns.boxplot(x=uic_dow["day_name"], y=uic_dow["rides"], order=day_order, color=CTA_BLUE, ax=axes[0])
3 sns.boxplot(x=wabash_dow["day_name"], y=wabash_dow["rides"], order=day_order, color=CTA_ORANGE, ax=axes[1])
4 axes[0].set_title("UIC-Halsted (student / commuter)")
5 axes[1].set_title("Washington/Wabash (downtown Loop)")
6 for ax in axes: ax.set_xlabel(""); ax.tick_params(axis="x", rotation=45)
7 plt.tight_layout()
```



Two Stations, Two Rhythms? (Monthly)

```
1 wabash_month = wabash.with_columns(pl.col("date").dt.strftime("%b").alias("month"))
2 fig, axes = plt.subplots(1, 2, figsize=(10, 3.09), sharey=True)
3 sns.boxplot(x=uic_month["month"], y=uic_month["rides"], order=month_order, color=CTA_BLUE, ax=axes[0])
4 sns.boxplot(x=wabash_month["month"], y=wabash_month["rides"], order=month_order, color=CTA_ORANGE, ax=axes[1])
5 axes[0].set_title("UIC-Halsted (student / commuter)")
6 axes[1].set_title("Washington/Wabash (downtown Loop)")
7 for ax in axes: ax.set_xlabel(""); ax.tick_params(axis="x", rotation=45)
8 plt.tight_layout()
```



Intuition

- **Weekday : weekend ratio**
 - the Loop has more weekend traffic (shopping, tourism, events, etc.)
- **Academic calendar**
 - UIC dips in January, summer, and December
 - the Loop's annual seasonal swing is weaker
- **History** — Washington/Wabash only opened in **Aug 2017**
 - (replaced Madison/Wabash and Randolph/Wabash)
 - its series is much shorter

More Forecasting Models

Forecasting Models: Overview

May be useful when working with more complex, intricate time series.

- **AR(p)** — predict from the last p observations
- **MA(q)** — predict from the last q forecast errors
- **ARMA(p, q)** — combines AR and MA; requires a stationary series
- **ARIMA(p, d, q)** — ARMA after d rounds of differencing; handles trend
- **SARIMA(p, d, q)(P, D, Q) m** — adds seasonal AR, differencing, MA

Bayesian Time Series

Why Bayesian?

- Classical models usually just give a **single point forecast**
 - a Bayesian model returns a **full posterior distribution** for every quantity
 - every estimate and prediction comes with a **credible interval**
- **Priors** let you encode what you already know
 - e.g. ridership can't be negative
 - weekends will be different
- Plays nicely with small data, missing days

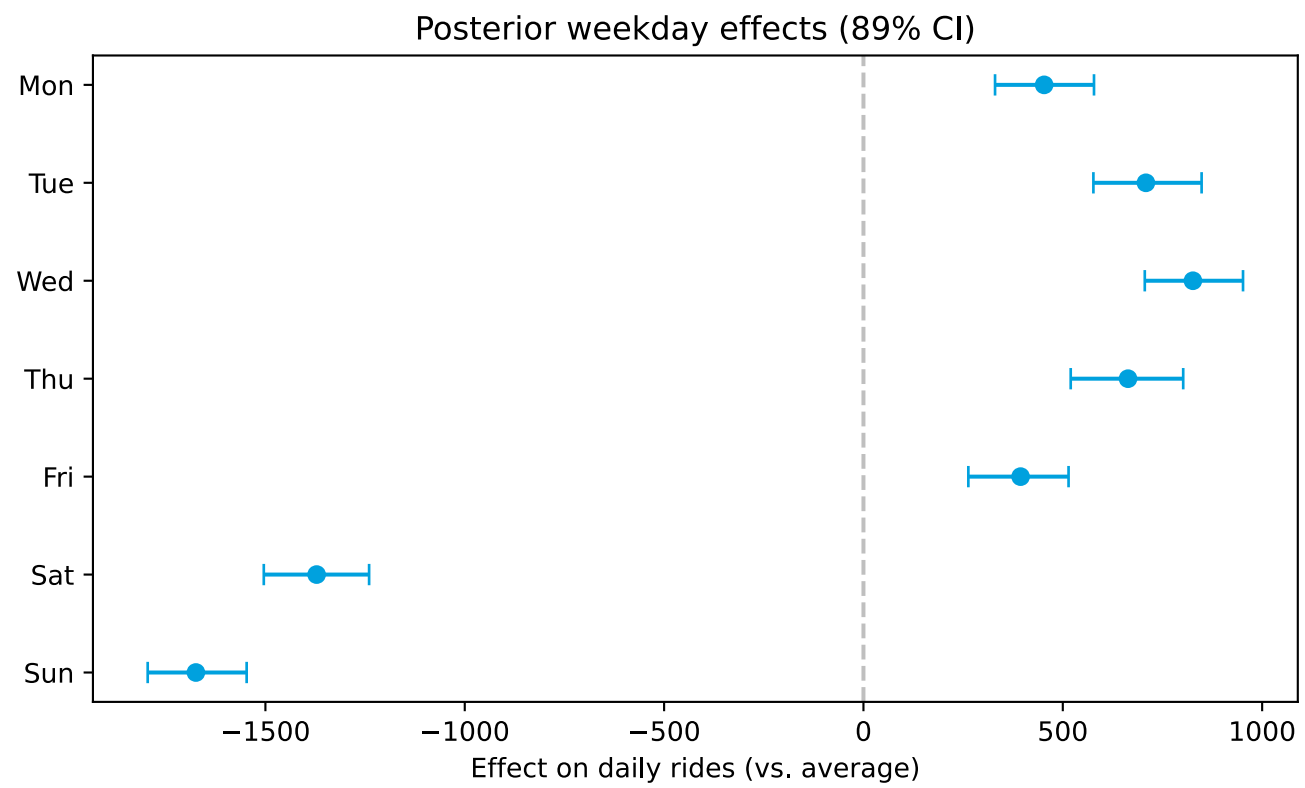
A Bayesian Ridership Model

Let's start by modeling each day as **intercept + weekday effect + month effect + noise**, and let the data update our beliefs.

```
1 import numpy as np
2 import pymc as pm
3
4 recent = uic.filter(pl.col("date") >= pl.date(2023, 1, 1)).with_columns(
5     (pl.col("date").dt.weekday() - 1).alias("dow"),      # 0=Mon ... 6=Sun
6     (pl.col("date").dt.month() - 1).alias("month"),      # 0=Jan ... 11=Dec
7 )
8 dow = recent["dow"].to_numpy()
9 month = recent["month"].to_numpy()
10 y = recent["rides"].to_numpy().astype(float)
11
12 with pm.Model() as model:
13     intercept = pm.Normal("intercept", mu=y.mean(), sigma=2000)
14     dow_eff = pm.ZeroSumNormal("dow_eff", sigma=1500, shape=7)
15     month_eff = pm.ZeroSumNormal("month_eff", sigma=1500, shape=12)
16     sigma = pm.HalfNormal("sigma", 2000)
17     mu = intercept + dow_eff[dow] + month_eff[month]
18     pm.Normal("obs", mu=mu, sigma=sigma, observed=y)
```


Posterior: Weekday Effects

```
1 post = idata.posterior
2 dow_s = post["dow_eff"].values.reshape(-1, 7)
3 means = dow_s.mean(0); lo, hi = np.percentile(dow_s, [5.5, 94.5], 0)
4 fig, ax = plt.subplots(figsize=(7, 4.3))
5 ax.errorbar(means, range(7), xerr=[means-lo, hi-means], fmt="o", color=CTA_BLUE, capsize=4)
6 ax.axvline(0, color="gray", linestyle="--", alpha=0.5)
7 ax.set_yticks(range(7)); ax.set_yticklabels(["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]); ax.invert_yaxis()
8 ax.set(xlabel="Effect on daily rides (vs. average)", title="Posterior weekday effects (89% CI)")
9 plt.tight_layout()
```



Using the Model to Predict

Fitting and Forecasting

Each posterior draw is one **plausible version of the world**.

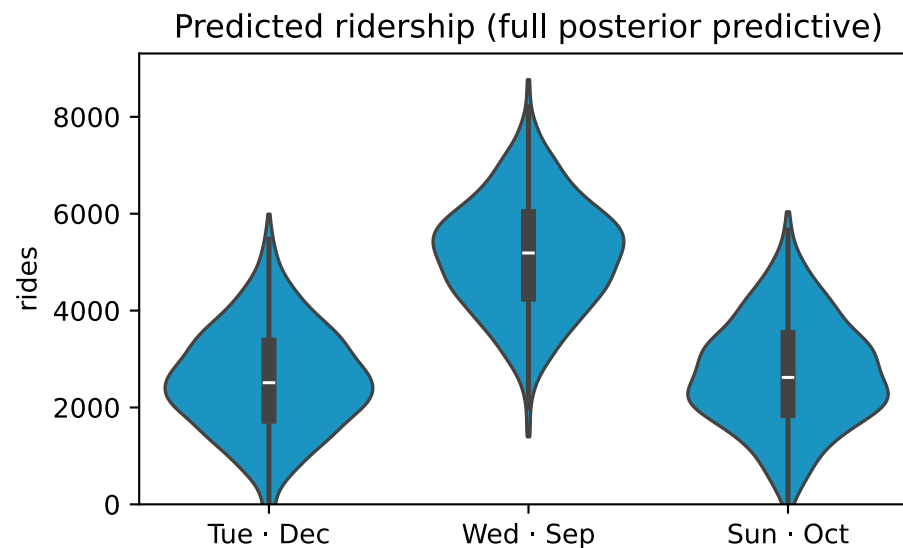
- To predict a new day:
 - push (**weekday, month**) through *every* draw (every version of the world)
- That gives a **distribution** of outcomes — not just a single number!
- The **spread** provides uncertainty
 - only use **mean** if asked 😊
- e.g. “how many rides on a *Tuesday in December?*” → let’s look!

Predicting a Given Day

```
1 intercept_s = post["intercept"].values.flatten()
2 month_s     = post["month_eff"].values.reshape(-1, 12)
3 sigma_s     = post["sigma"].values.flatten()
4 rng = np.random.default_rng(0)
5
6 def predict(day, mon):
7     mu = intercept_s + dow_s[:, day] + month_s[:, mon]
8     return rng.normal(mu, sigma_s)
```

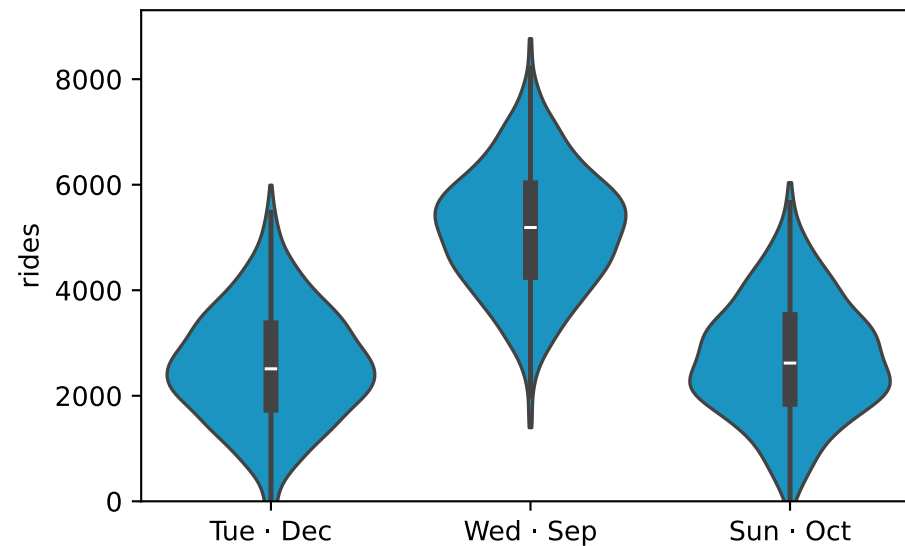
Predicting a Given Day

```
1 scenarios = {"Tue · Dec": (1, 11), "Wed · Sep": (2, 8), "Sun · Oct": (6, 9)}
2 frames = []
3 for k, v in scenarios.items():
4     r = predict(*v)
5     frames.append(pl.DataFrame({"scenario": [k] * len(r), "rides": r}))
6 draws = pl.concat(frames)
7 ax = sns.violinplot(x=draws["scenario"], y=draws["rides"], color=CTA_BLUE, cut=0)
8 ax.set_ylim(bottom=0); ax.set_xlabel("")
9 ax.set_title("Predicted ridership (full posterior predictive)")
10 plt.tight_layout()
```



Reading the Predictions

- **Tue in December:** $\approx 2,530$ rides, 89% CI: 480 – 4,620
- **Wed in September:** $\approx 5,150$ rides, 89% CI: 2,950 – 7,260
- **Sun in October:** $\approx 2,680$ rides, 89% CI: 610 – 4,860
- Intervals are wide (lots of **day-to-day noise** remains after weekday, month)



Visualizing Intervals

Ways to Visualize Intervals

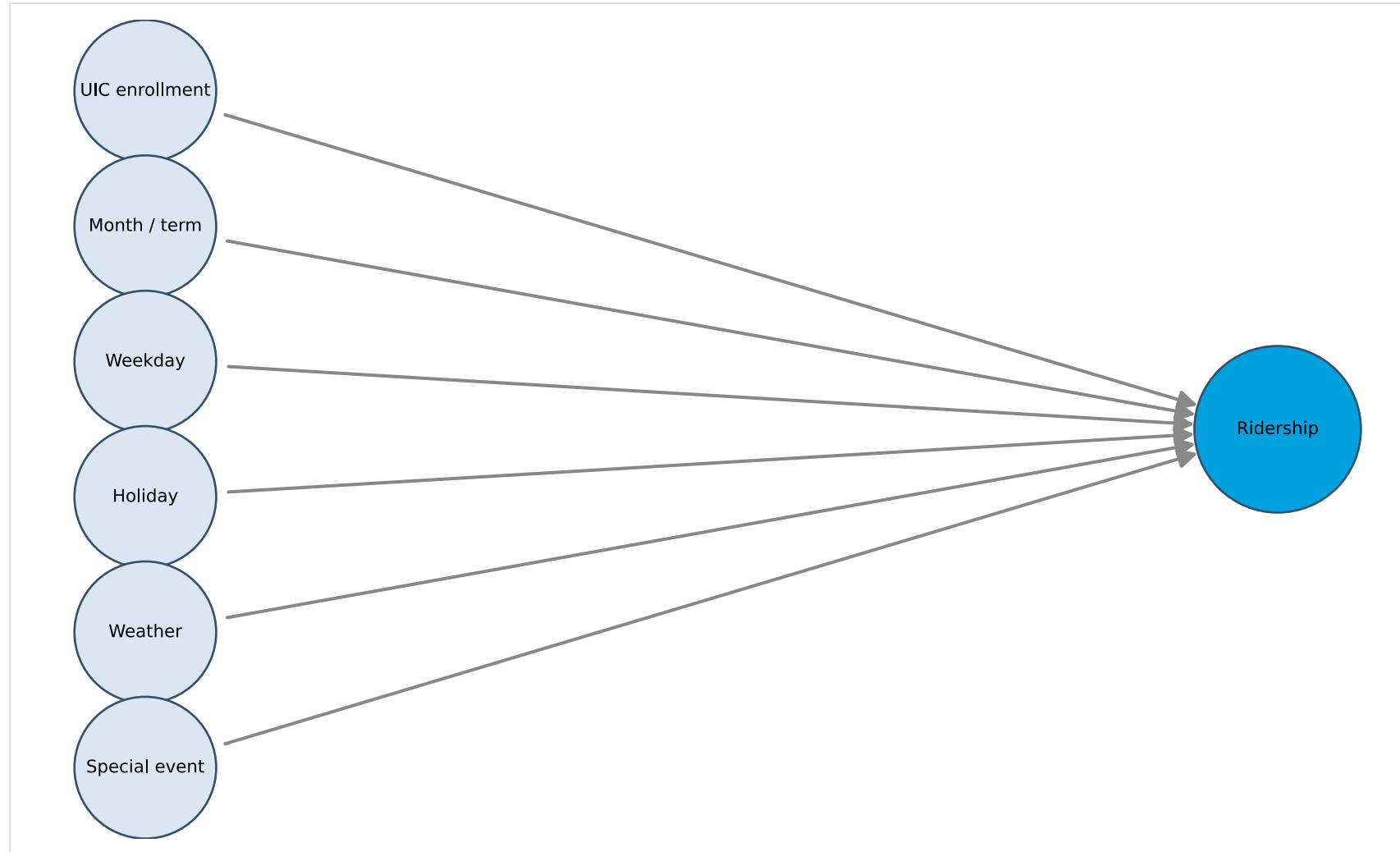
- TK
- TK1
- TK2

Another Approach: Ignore Time

So far, the data was essentially the model. But ridership responds to factors (dare we say **causes??**), which can be identified and measured.

- Treat each day as an independent **observation** (a “unit”), not a step in a sequence.
- Predict from **covariates**: enrollment, weekday, holiday, special event, weather, etc...
 - What else comes to mind?
- This is a **causal / regression** framing
 - a preview of how we'll model

A Simple Causal DAG for Ridership

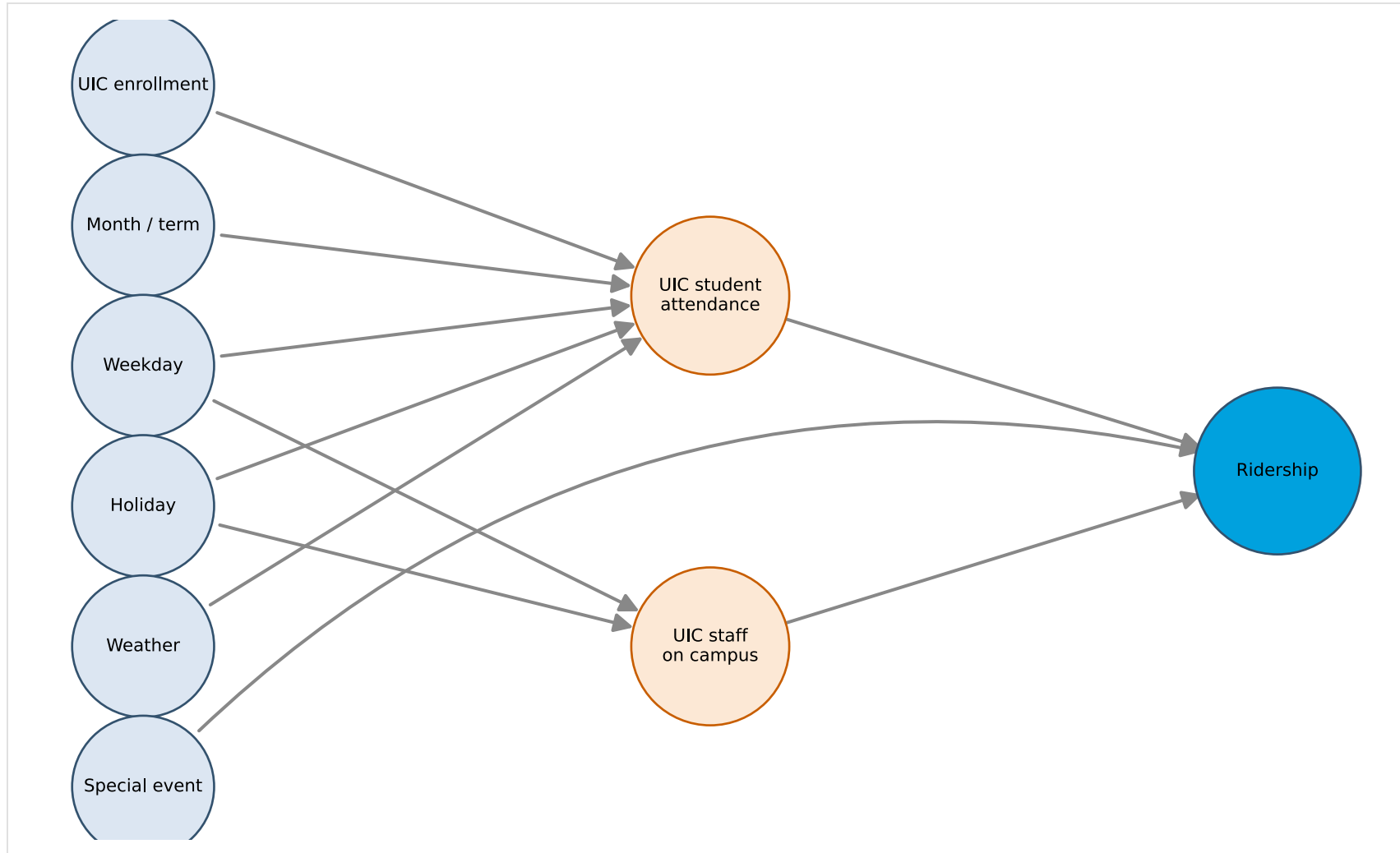


A Causal DAG for Ridership

What drives daily ridership at a station? Not always directly — some causes work through **intermediate factors**.

- **UIC enrollment + month/term + weather** → **UIC student attendance**
- **Weekday + holiday** → student attendance, **UIC staff on campus**
 - note the *academic calendar* affects students but **not** staff
- **Special event** like concerts, games, speakers → directly boosts ridership (could also do this as a hierarchy)
- **Ridership** ← student attendance + staff on campus + special events + etc.

A Hierarchical Causal DAG for Ridership



Sources

1. GitHub source: <https://github.com/jackbandy/data-science-fun/blob/main/docs/slides/week6.qmd>.
2. Last modified and compiled June 22, 20:11h Central (Chicago) time.
3. Time series decomposition: Jason Brownlee, *How to Decompose Time Series Data into Trend and Seasonality*, Machine Learning Mastery.
4. Airline passengers dataset:
 - Box, G.E.P., Jenkins, G.M. (1976). *Time Series Analysis: Forecasting and Control*. Holden-Day. ISBN 978-0-8162-1104-3. ([Internet Archive scan](#)) ([5th edition via Wiley](#))
 - Dataset in R's base `datasets` package: `AirPassengers` — official R documentation with column definitions and citation
 - Rdatasets mirror (CSV download): <https://vincentarelbundock.github.io/Rdatasets/doc/datasets/AirPassengers.html>
5. CTA ridership data: [Chicago Data Portal — CTA Ridership: Station Entries — Daily Totals](#).
6. Slides developed using materials from [Elena Zheleva](#) and [Gonzalo Bello Lander](#), the Berkeley DS 100 team, Marine Carpuat, and Brian Ziebart.
7. Slide deck built with [Quarto](#) revealjs.
8. Title font is Big Shoulders; Body font is [Libre Franklin](#).

Appendix

ACF and PACF

Use ACF and PACF plots to choose p and q from data. Two tools for choosing model order from data:

- **ACF (Autocorrelation Function)** — correlation between y_t and y_{t-k} for each lag k . Use to choose MA order q : ACF “cuts off” after lag q .
- **PACF (Partial Autocorrelation Function)** — correlation at lag k after removing the effects explained by shorter lags. Use to choose AR order p : PACF “cuts off” after lag p .

Reading ACF and PACF Plots

Each bar = correlation at that lag; shaded band = 95% confidence interval.

- Bars **within** the band → not significant (treat as zero)
- **ACF decays slowly** → series may not be stationary — try differencing first
- **ACF cuts off** sharply after lag q → suggests **MA(q)**
- **PACF cuts off** sharply after lag p → suggests **AR(p)**
- **Both decay gradually** → suggests **ARMA(p, q)** — compare with AIC / BIC

ACF and PACF: Airline Passengers

```
1 from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
2
3 fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(7, 4.3))
4 plot_acf(values, lags=30, ax=ax1, title="ACF — Airline Passengers")
5 plot_pacf(values, lags=30, ax=ax2, title="PACF — Airline Passengers")
6 plt.tight_layout()
```

